10.5.1 Classifications of Colloidal solution

Probably the most important colloidal systems have dispersed phase as solid and the dispersion medium as liquid.

If the dispersion medium considered is water, then the colloids are referred as hydrosols or aquasols.

If the dispersion medium is an alcohol, the colloid is termed as alcosol, and if benzene is the dispersion medium, it is called as benzosol.

One more type of classification is based on the forces acting between the dispersal phase and dispersion medium.

In lyophillic colloids definite attractive force or affinity exists between dispersion medium and dispersed phase. Examples: sols of protein and starch. They are more stable and will not get precipitated easily. They can be brought back to colloidal solution even after the precipitation by addition of the dispersion medium.

In a lyophobic colloids, no attractive force exists between the dispersed phase and dispersion medium. They are less stable and precipitated readily, but can not be produced again by just adding the dispersion medium. They themselves undergo coagulation after a span of characteristic life time.

They are called irreversible sols

examples: sols of gold, silver, platinum and copper.

The following table lists the types of colloids based on the physical states of dispersed phase and dispersion medium.

Classification of colloids based on the physical state of dispersed phase and dispersion medium.

S.No.	Dispersion medium	Dispersed phase	Name of the colloid	Examples
1.	Gas	Liquid	Liquid Aerosol	Fog Aerosol spray
2.	Gas	Solid	Solid Aerosol	Smoke, Air pollutants like fumes, dust.
3.	Liquid	Gas	Foam	Whipped cream, Shaving cream, Soda water, Froth.
4.	Liquid	Liquid	Emulsion	Milk, Cream, Mayonnaise
5.	Liquid	Solid	Sol	Inks, Paints, colloidal gold.

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